

Computer Network Lab

Assignment - Designing an Enterprise Network using Packet Tracer

A. Problem Statement

In this project, you will implement the design of an enterprise network (i.e. **UIU**) using the Cisco Packet Tracer software. You must configure the network design according to the requirements given in the “Project Requirements” section below. After designing the network, you should also submit a **1-3 page report** regarding what you have learned and demonstrated to complete this project.

B. Project Requirements

Your project should have the following features (Marks for implementing each feature is given in the right-hand side):

- a. Your enterprise network should have at least **30 hosts** (Could be PCs, Laptops, Servers) and at least **15 among them must be PCs** 3
- b. Your enterprise network should have at least **6 routers** and **6 switches/hubs**. You can use any router model for creating the network and use any kind of connection to connect all the devices. However, **you cannot attach the 6 routers in a line**. You should connect the routers in a way that mimics real-life settings of enterprise networks 3
- c. Your enterprise network should be **divided into two parts** - one **private network** and one **public network**. You must use either **static or dynamic NAT** to connect the two networks together. 2
- d. You should create subnets for your enterprise network as per your wish, if your full enterprise network works correctly. Your submitted project must be fully operational and you should be able to ping all devices from each of the PCs 2
- e. You must use **RIP**, **OSPF**, or static routing to create the routing table for each of the routers. You can use the CLI to set them up. 2
- f. You must use **DHCP to set the IP addresses of all PCs in the network**. 2
- g. Remember to test if your enterprise network is working correctly
- h. Make sure all the devices added to your network are organized cleanly, the display and hostnames of all of the devices are set in an orderly fashion, and subnets are shown and marked neatly. 3

C. Report Requirements

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Your report should contain the following points:

- a. A short **50-word paragraph** on what you implemented in Packet Tracer
- b. **IOS commands that you had to use to set up the routers**
- c. Any challenges that you encountered while implementing the enterprise network

D. Submission Requirement

You have to submit the Packet Trace **.pkt** file and the report file (in **.pdf**) in eLMS.

E. Submission Deadline

N.B: This deadline is tentative and may change with time

- a. For Section A - Jan 29, 2025
- b. For Section G - Jan 29, 2025